





How can we determine $S_z(\gamma)$?

- ★ Optical methods do not apply, since γ is too energetic, of order MeV.
- ★ Absorption by polarized electrons **in matter** is spin selective

$$\overrightarrow{\gamma} + \overleftarrow{e} \rightarrow \overrightarrow{e}$$

Photon absorbed.
Electron's spin flipped

The electron spin direction
can be reversed by an
external magnetic field.

$$\overrightarrow{\gamma} + \overrightarrow{e} \rightarrow \overrightarrow{e}$$

Photon not absorbed. Angular
momentum conservation would
require $S_z(e) = 3/2$, which is
impossible.

How can we measure $S_z(\gamma)$?